

Course Title: Software Development III

Course Code: CSE 4104

Project Name: FuturePath AI-An AI-Powered Internship Matching and Career Readiness Platform

Submitted By-
Farhana Zaman Ahona
ID: 11230321365
Sumaia
ID: 11230321253
Sumaiya Hafiz
ID: 11230321247

Submitted To-
Md. Riaz Mahmud
Assistant Professor
Department of CSE, NUBTK

Project Proposal

1.Project Title

FuturePath AI – An AI-Powered Internship Matching and Career Readiness Platform

2.Problem Statement

University students and fresh graduates often struggle to find suitable internship opportunities that match their skills, education, and career interests. Many students also do not know which skills they need to improve to become eligible for their desired internships.

- **What problem currently exists?**

  Many university students struggle to find suitable internships based on their skills, education, and interests. They also face difficulties in creating a good CV, identifying missing skills, preparing for interviews, and choosing the right career path. As a result, many students miss valuable opportunities due to a lack of proper guidance.

- **Why is the problem important?**

  Internships help students gain practical experience and prepare for professional life. However, many students lack the skills and confidence required by companies. Proper internship guidance, skill development, and career preparation are important to help students become job-ready and achieve their career goals.

- **Why are existing solutions insufficient?**

  Most existing platforms mainly focus on listing internship opportunities. They provide limited support for CV analysis, skill gap identification, interview preparation, and personalized learning roadmaps. Students often need multiple platforms to manage their internship search and career development.

- **Why is AI relevant for solving the problem?**

  AI can analyze students' CVs, skills, education, and interests to recommend suitable internships and job roles. It can identify skill gaps, suggest learning resources, create personalized learning roadmaps, and provide career guidance and intervie preparation. This makes the internship search process smarter and more personalized.

3.Objectives

- To help students find suitable internship opportunities.
- To analyze students' CVs and identify areas for improvement.
- To identify skill gaps based on internship requirements.
- To provide personalized learning roadmaps.
- To offer AI-based career guidance and interview preparation.
- To make internship applications easier to manage and track.

4. Proposed Features

1. AI-Powered Internship Matching: Matches students with suitable internships based on their skills and interests.

2. AI-Based CV Analysis: Analyzes students' CVs and identifies missing or weak information.

3. Smart Skill Gap Analysis: Finds the skills students need to improve for their desired careers.

4. Personalized Learning Roadmap: Provides a learning plan to help students develop the required skills.

5. AI Career Guidance Assistant: Gives career advice and answers students' career-related questions.

6. Internship Application Tracking: Allows students to track their internship application status.

7. Student Profile Management: Allows students to create and update their personal, academic, and skill information.

8. Internship Search and Filtering: Helps students find internships based on location, skills, and other preferences.

9. Internship Eligibility Checker: Checks whether students meet the requirements for specific internships.

10. AI Interview Preparation: Helps students prepare for interviews with AI-generated questions and tips.

11. Personalized Job Role Suggestions: Suggests suitable job roles based on students' skills and interests.

12. Skill Development Resource Suggestions: Recommends useful courses, tutorials, and learning materials.

13. Application Deadline Notifications: Notifies students about upcoming internship application deadlines.

5. Technology Stack

Frontend:

- React.js

Backend:

- Python (Django)

Database:

- PostgreSQL

AI Integration:

- Google Gemini API

Code Editor:

- Visual Studio Code

6.Project planning

The project will be developed step by step to ensure all important features work properly.

Phase	**Activities**
Phase 1: Planning	Select project goals, features, and requirements.
Phase 2: UI/UX Design	Design the website layout, dashboard, and user interface
Phase 3: Database Design	Create the database for student profiles, internships, CVs, and applications.
Phase 4: Frontend Development	Develop the website using HTML, CSS, JavaScript, and Bootstrap.
Phase 5: Backend Development	Develop login, registration, profile management, internship search, and application tracking using PHP and MySQL.
Phase 6: AI Integration	Implement AI-based CV analysis, internship matching,

	skill gap analysis, and career guidance.
Phase 7: Testing	Test all features, find errors, and fix them.
Phase 8: Final Delivery	Complete the project, prepare documentation, and present the final system.

7. Expected Outcome

FuturePath AI will provide a platform where students can discover suitable internships and prepare for their future careers.

Users will receive personalized internship recommendations, CV improvement suggestions, skill gap reports, and learning roadmaps. The application will also help students manage their internship applications and prepare for interviews.

The AI integration will make the platform more personalized and useful by providing recommendations based on individual student profiles.